

On the Operator Alpha-Ratio and Alpha-Reiter Problems

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Source Problems

The source paper is N. Monod, *Fixed points in convex cones*, arXiv:1701.05537. Near the end of the paper, after discussing ℓ^p -variants of the f -ratio and f -Reiter properties, the author introduces the following Hilbert-space analogues. A unitary representation π of a group G on a Hilbert space V has the operator α -ratio property, for a continuous operator $\alpha \neq 0$, if for every finite $S \subseteq G$ and every $\varepsilon > 0$ there is $v \in V$ such that

$$\|\alpha\pi(s)v\| < (1 + \varepsilon)\|\alpha v\| \quad (s \in S).$$

It has the stronger operator α -Reiter property if one can choose v such that

$$\|\alpha(\pi(s)v - v)\| < \varepsilon\|\alpha v\| \quad (s \in S).$$

Problems 47 and 48 ask to clarify these properties, especially for projections α .

straightforward verification where J , u and ϵ are to be replaced with appropriate powers. The same is true for the f -Reiter property using basic L^p inequalities.

The particular case of $p = 2$ suggests the following:

We say that a unitary representation of a group G on a Hilbert space V has the **operator α -ratio property** for a continuous operator $\alpha \neq 0$ of V if for every finite set $S \subseteq G$ and every $\epsilon > 0$, there is $v \in V$ with $\|\alpha(sv)\| < (1 + \epsilon)\|\alpha(v)\|$ for all $s \in S$. Likewise, the representation has the stronger **operator α -Reiter property** if we can choose v with $\|\alpha(sv - v)\| < \epsilon\|\alpha(v)\|$ for all $s \in S$.

Problem 47. Clarify under which circumstances the operator α -ratio and α -Reiter properties hold.

Returning closer to supramenability:

Problem 48. Characterise the unitary representations that satisfy the operator α -ratio or α -Reiter property for all non-zero projections α in V .

The following theorem gives a direct clarification. The ratio property is automatic as stated; the Reiter property is exactly the absence of a relative spectral gap for the seminorm induced by α .

Idea of the Proof

The ratio condition only asks that the α -size of finitely many unitary translates not exceed the original α -size by much. A vector which almost attains the operator norm of α already has this property, because unitarity gives $\|\pi(s)v\| = \|v\|$ and hence $\|\alpha\pi(s)v\| \leq \|\alpha\| \|v\|$.

For Reiter, there is no such norm-attainment trick. The obstruction is exactly a Poincare inequality measured by α : a finite set S prevents α -Reiter vectors precisely when the α -movement under S is bounded below by the α -size of the vector.

Theorem 1. *Let $\pi : G \rightarrow \mathcal{U}(V)$ be a unitary representation on a Hilbert space and let $0 \neq \alpha \in B(V)$.*

1. *The representation has the operator α -ratio property.*
2. *The representation has the operator α -Reiter property if and only if, for every finite $S \subseteq G$,*

$$\inf_{\alpha v \neq 0} \frac{\sum_{s \in S} \|\alpha(\pi(s)v - v)\|^2}{\|\alpha v\|^2} = 0.$$

Equivalently, for no finite $S \subseteq G$ does there exist a constant $c > 0$ such that

$$\sum_{s \in S} \|\alpha(\pi(s)v - v)\|^2 \geq c \|\alpha v\|^2 \quad \text{for all } v \in V.$$

3. *If α is bounded below, then the operator α -Reiter property is equivalent to the ordinary existence of almost invariant vectors for π .*

Proof. Fix a finite set $S \subseteq G$ and $\varepsilon > 0$. Since $\alpha \neq 0$, choose a unit vector v with

$$\|\alpha v\| > \frac{\|\alpha\|}{1 + \varepsilon}.$$

For every $s \in S$, unitarity gives

$$\|\alpha\pi(s)v\| \leq \|\alpha\| < (1 + \varepsilon)\|\alpha v\|.$$

This proves the operator α -ratio property.

For the Reiter assertion, the displayed infimum is zero exactly when, for every $\delta > 0$, there is v with $\alpha v \neq 0$ and

$$\sum_{s \in S} \|\alpha(\pi(s)v - v)\|^2 < \delta^2 \|\alpha v\|^2.$$

This implies the defining Reiter inequalities for all $s \in S$ after choosing $\delta = \varepsilon$. Conversely, the defining inequalities imply the displayed sum estimate with $\delta^2 = |S|\varepsilon^2$. Thus the infimum formulation and the original definition are equivalent. The second formulation is just the negation of the existence of a positive lower bound for the same Rayleigh quotient.

Finally suppose that α is bounded below: there are constants $0 < m \leq M < \infty$ such that

$$m\|w\| \leq \|\alpha w\| \leq M\|w\| \quad (w \in V).$$

If π has almost invariant vectors, then for a given finite S choose $v \neq 0$ with

$$\|\pi(s)v - v\| < \varepsilon m M^{-1} \|v\| \quad (s \in S).$$

It follows that

$$\|\alpha(\pi(s)v - v)\| \leq M\|\pi(s)v - v\| < \varepsilon m\|v\| \leq \varepsilon\|\alpha v\|.$$

Thus α -Reiter holds. Conversely, if α -Reiter holds, choose v with

$$\|\alpha(\pi(s)v - v)\| < \varepsilon m M^{-1}\|\alpha v\| \quad (s \in S).$$

Then

$$m\|\pi(s)v - v\| \leq \|\alpha(\pi(s)v - v)\| < \varepsilon m M^{-1}\|\alpha v\| \leq \varepsilon m\|v\|,$$

so $\|\pi(s)v - v\| < \varepsilon\|v\|$ for all $s \in S$. Hence π has almost invariant vectors. $\square$

It is sometimes useful to write the criterion in operator form. If

$$A_S = \sum_{s \in S} (\pi(s) - I)^* \alpha^* \alpha (\pi(s) - I), \quad B = \alpha^* \alpha,$$

then α -Reiter fails exactly when $A_S \geq cB$ for some finite S and some $c > 0$.

Corollary 1 (Finite-rank projections). *Let P be an orthogonal projection with finite-dimensional range W . Then π has the operator P -Reiter property if and only if, for every finite $S \subseteq G$, there exists $v \in V$ such that*

$$Pv \neq 0 \quad \text{and} \quad P(\pi(s)v - v) = 0 \quad (s \in S).$$

Proof. For fixed S , consider the linear map

$$L : V \longrightarrow W \oplus W^S, \quad L(v) = (Pv, (P(\pi(s)v - v))_{s \in S}).$$

The range of L is a subspace of the finite-dimensional space $W \oplus W^S$, hence is closed.

If the exact condition holds, then the P -Reiter inequalities hold with zero left hand side for this finite S . Conversely, if P -Reiter holds, choose v_n with $\|Pv_n\| = 1$ and

$$\sum_{s \in S} \|P(\pi(s)v_n - v_n)\|^2 \longrightarrow 0.$$

After passing to a subsequence, compactness of the unit sphere of W gives $Pv_n \rightarrow w$ with $\|w\| = 1$. Hence $(w, 0)$ belongs to the closure of $L(V)$, and therefore to $L(V)$ itself. This gives a vector v satisfying the exact condition. $\square$

Corollary 2 (Rank-one projections). *Let P_ξ be the orthogonal projection onto the line spanned by a non-zero vector ξ . Then π has the operator P_ξ -Reiter property if and only if, for every finite $S \subseteq G$,*

$$\xi \notin \text{span}\{\pi(s)^{-1}\xi - \xi : s \in S\}.$$

Proof. By Corollary 1, for each finite S we need a vector v such that $\langle v, \xi \rangle \neq 0$ and

$$\langle \pi(s)v - v, \xi \rangle = 0 \quad (s \in S).$$

The latter equations say exactly that v annihilates the finite-dimensional space

$$M_S = \text{span}\{\pi(s)^{-1}\xi - \xi : s \in S\}.$$

Such a v with $\langle v, \xi \rangle \neq 0$ exists precisely when $\xi \notin M_S$. $\square$

Consequences and Limitations

The operator α -ratio property, as formulated in the source problem, is therefore vacuous for bounded non-zero operators: it holds for every unitary representation of every group.

The operator α -Reiter property is genuinely representation-theoretic. For $\alpha = I$ it is exactly the usual almost-invariant-vector condition. In particular, for the left regular representation it is equivalent to amenability of the group. More generally, when α is bounded below, Theorem 1 shows that no new condition is obtained beyond ordinary almost invariance.

For projections with finite-dimensional range, the approximate condition collapses to the exact finite-dimensional condition in Corollary 1. This gives a concrete answer to the finite-rank projection subcase of Problem 48. The packet does not claim to classify all infinite-rank projections, nor does it address the separate open problems about products, supramenability, or finitely generated groups of exponential growth with the fixed-point property for cones.

Novelty Check

The run indexes were searched for arXiv:1701.05537 and for phrases including `operator alpha-ratio`, `operator alpha-Reiter`, `fixed points in convex cones`, `conical fixed-point property`, and `supramenability`. No existing packet for these endpoint operator questions was found. Web searches for exact phrases such as `"operator alpha-ratio"`, `"operator alpha-Reiter"`, and variants including `"Fixed points in convex cones"` did not return a later explicit answer to Problems 47 or 48. The result here is elementary, but it appears to clarify a point left open in the source problem statement.

References

- [1] N. Monod, *Fixed points in convex cones*, arXiv:1701.05537, 2017.